Design & Analysis of Algorithms_2024_PYQ

Group-A (Very Short Answer Type Question)

Answer any ten of the following: [1 x 10 = 10]

1. What is the appropriate data structure for Prim's Minimum Spanning Tree algorithm?
2. What is represented by the asymptotic notation O(1)?
3. Time complexity for recurrence relation T(n)=2 T(n/2) + n is _____.
4. The notation Ω(n) is the formal way to express the _____ bound of an algorithm's running time.
5. Fractional knapsack problem is solved most efficiently by which algorithm design technique?
6. What is the appropriate data structure for Depth First Search algorithm?
7. Which algorithm is used to solve a maximum flow problem?
8. How many solutions are there for 8 queens' problem on 8*8 board?
9. Time complexity of Heap sort with n items is _____.
10. The 0-1 Knapsack problem can be solved using Greedy algorithm – state True or False.
11. Which algorithm is used to solve the single source shortest path problem in a graph with negative edge weights?
12. Given items as {value, weight} pairs {{40,20},{30,10},{20,5}}. The capacity of knapsack is 20. Find the maximum value output assuming items to be divisible.

Group-B (Short Answer Type Question)

Answer any three of the following: [5 x 3 = 15]

2. What are the differences between Dijkstra's algorithm and Bellman-Ford algorithm? Compare their time complexities. [5]
3. Analyze the time complexity of quick sort algorithm for best case and worst-case using recurrence relations. [5]
4. Write the Floyd-Warshal Algorithm for all-pairs shortest path problem. What will be the time complexity? [5]

5. For the given set of 5 items and the knapsack capacity of 10 kg, find the maximum profit. You are allowed to take fractional amount of any item. Weights, W_i = {3, 3, 2, 5, 1}; Profits, P_i = {10, 15, 10, 12, 8} [5]

6. Write short note on Randomized algorithms. [5]

Group-C (Long Answer Type Question)

Answer any three of the following: [15 x 3 = 45]

7. (a) Find the minimum number of multiplications required for the following matrix chain multiplication using Dynamic programming:

$$A(10 \times 20) * B(20 \times 50) * C(50 \times 1) * D(1 \times 100)$$ [7]

(b) Write the algorithm for single source shortest path (negative edge weights allowed). [5]

(c) Discuss the advantages and drawbacks of Dynamic Programming. [3]

8. (a) What are the features of any Algorithm? [3]

(b) Discuss the different Asymptotic notations and their importance. [5]

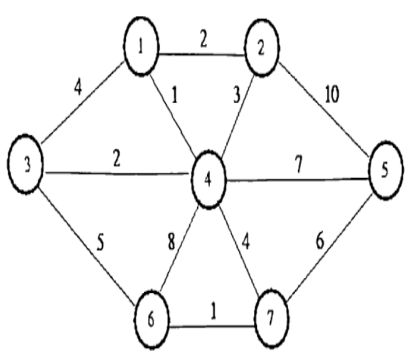
(c) Write the recurrence relation for computing the time complexity of 'Towers of Hanoi' problem. Also solve the recurrence relation to get the time complexity. [7]

9. (a) Write the greedy algorithm for job sequencing with deadline. [4]

(b) Using greedy method, find an optimal solution to the problem of job sequencing with deadline where $n = 4$, $(p_1, p_2, p_3, p_4) = (100,10,5,27)$ and $(d_1, d_2, d_3, d_4) = (2,1,2,1)$. [5]

(c) Define spanning tree. [2]

(d) Using Prim's algorithm generate the MST from the given graph. [4]



10. (a) Write and explain the classes P, NP, NP-hard and NP-complete. [3]

(b) What is reducibility? [5]

(c) If any NP-complete problem can be solved in polynomial time then P=NP – explain. [5]

11. (a) What is a Flow Network? What are the constraints of flow in a network? [5]

(b) What is an augmenting path in a flow network? Explain residual capacity – how it is calculated. [5]

(c) Write the Ford-Fulkerson Algorithm for finding maximum flow. [5]